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7.3C – Iteration 6

Location.cs

```
namespace SwinAdventure;


public class Location : GameObject, IHaveInventory
{
   private Inventory _inventory;
   public Location(string name, string desc) : base(new string[] { name.ToLower() }, name,
desc)
   {
      _inventory = new Inventory();
   }
   public GameObject Locate(string id)
   {
      if (AreYou(id)) { return this; }
      return Inventory.Fetch(id);
   }
   public Inventory Inventory { get { return _inventory; } }
   public override string FullDescription
   {
      get
      {
      return $"You are at {base.Name}\nThis is a {base.FullDescription}\nIn this {base.Name},
you can see {Inventory.ItemList}";
```

```
        }
    }
}
```

GameObject.cs

```
namespace SwinAdventure
{
    public abstract class GameObject : IdentifiableObject
    {
        private string _description;
        private string _name;
        public GameObject(string[] ids, string name, string desc) : base(ids)
        {
            _name = name;
            _description = desc;
        }
        public string Name
        {
            get { return _name; }
        }
        public string ShortDescription
        {
            get { return _name + " (" + FirstID + ")"; }
        }
        public virtual string FullDescription
        {
            get { return _description; }
```

```
        }
    }
}
```

Player.cs

```
using System.Reflection.Metadata.Ecma335;

namespace SwinAdventure
{
    public class Player : GameObject, IHaveInventory
    {
        private Inventory _inventory;
        private Location _location;
        public Player(string name, string desc) : base(new string[] { "me", "inventory" }, name, 
desc)
        {
            _inventory = new Inventory();
            _location = null;
        }
        public GameObject Locate(string id)
        {
            if (AreYou(id)) return this;
            GameObject item = Inventory.Fetch(id);
            if (item != null) return item;
            if (Location != null) return Location.Locate(id);
            return null;
        }
```

```
  public override string FullDescription
      {
          get
          {
              return $"You are {Name}, {base.FullDescription}.\nYou are
carrying:\n{Inventory.ItemList}";
          }
      }
      public Inventory Inventory
      {
          get { return _inventory; }
      }
      public Location Location
      {
          get { return _location; }
          set { _location = value; }
      }
  }
}
```

LookCommand.cs

```
namespace SwinAdventure;


public class LookCommand : Command
{


  public LookCommand() : base(new string[] { "look" }) { }
```

```
public override string Execute(Player p, string[] text)
{

   if (text[0].ToLower() != "look")
   {
      return "Error in look input";
   }
   if(text.Length== 1 && text[0].ToLower() == "look")
   {
      return p.Location.FullDescription;
   }
   if (text.Length != 3 && text.Length != 5)
   {
      return "I don't know how to look like that";
   }
   if (text[1].ToLower() != "at")
   {
      return "What do you want to look at?";
   }
   if (text.Length == 5 && text[3].ToLower() != "in")
   {
      return "What do you want to look in?";
   }
   string itemId = text[2];
   IHaveInventory container;
```

```
    if (text.Length == 3)
    {
        container = p;
    }
    else
    {
        string containerId = text[4];
        container = FetchContainer(p, containerId);
        if (container == null)
        {
            return $"I can't find the {containerId}";
        }
    }
    return LookAtIn(itemId, container);
}

private IHaveInventory FetchContainer(Player p, string containerId)
{
    GameObject obj = p.Locate(containerId);
    return obj as IHaveInventory;
}
private string LookAtIn(string thingId, IHaveInventory container)
{
    GameObject item = container.Locate(thingId);
    if (item == null)
    {
```

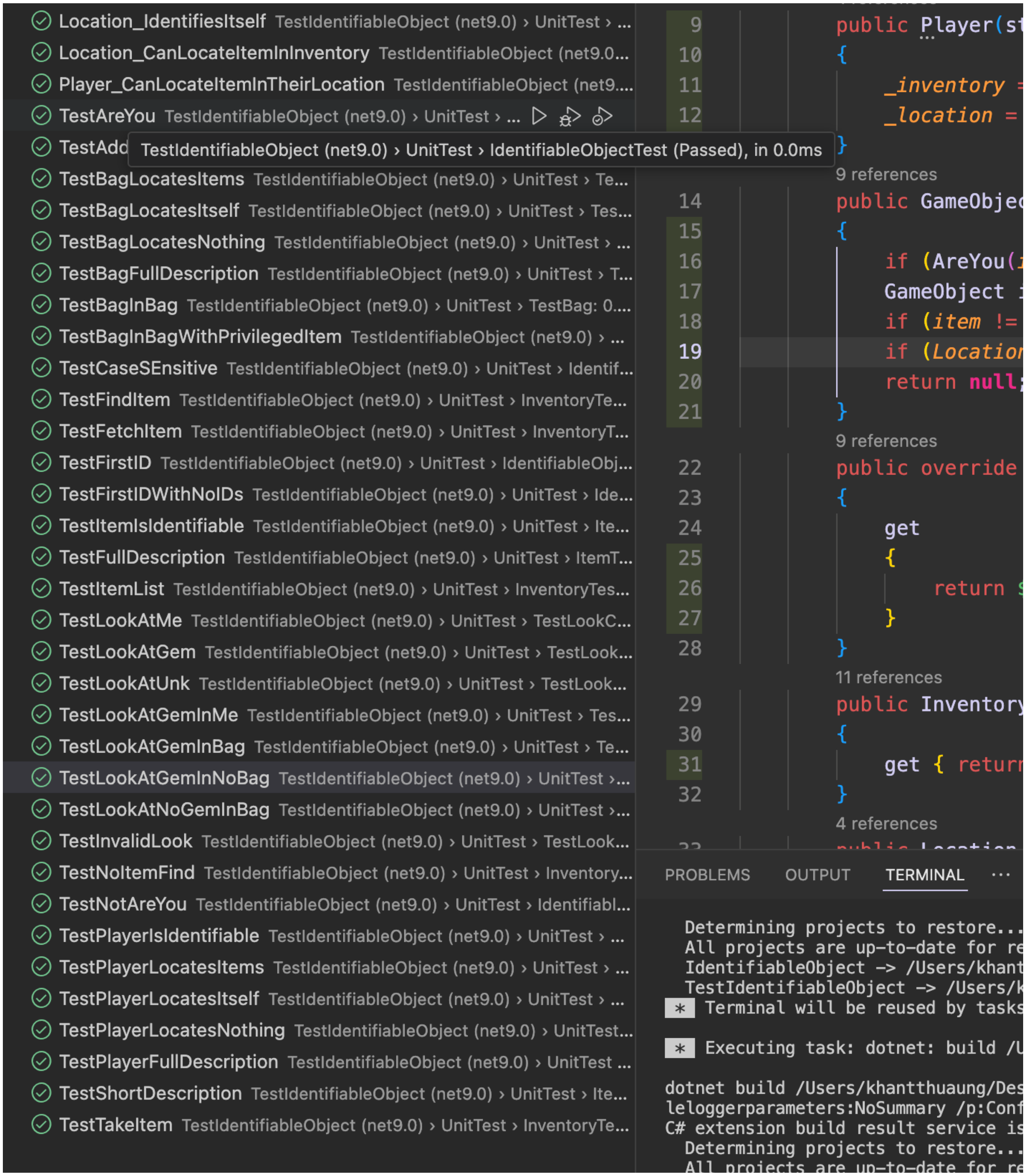
```
            return $"I can't find the {thingId}";
        }
        return item.FullDescription;
    }
}
```

TestLocation.cs

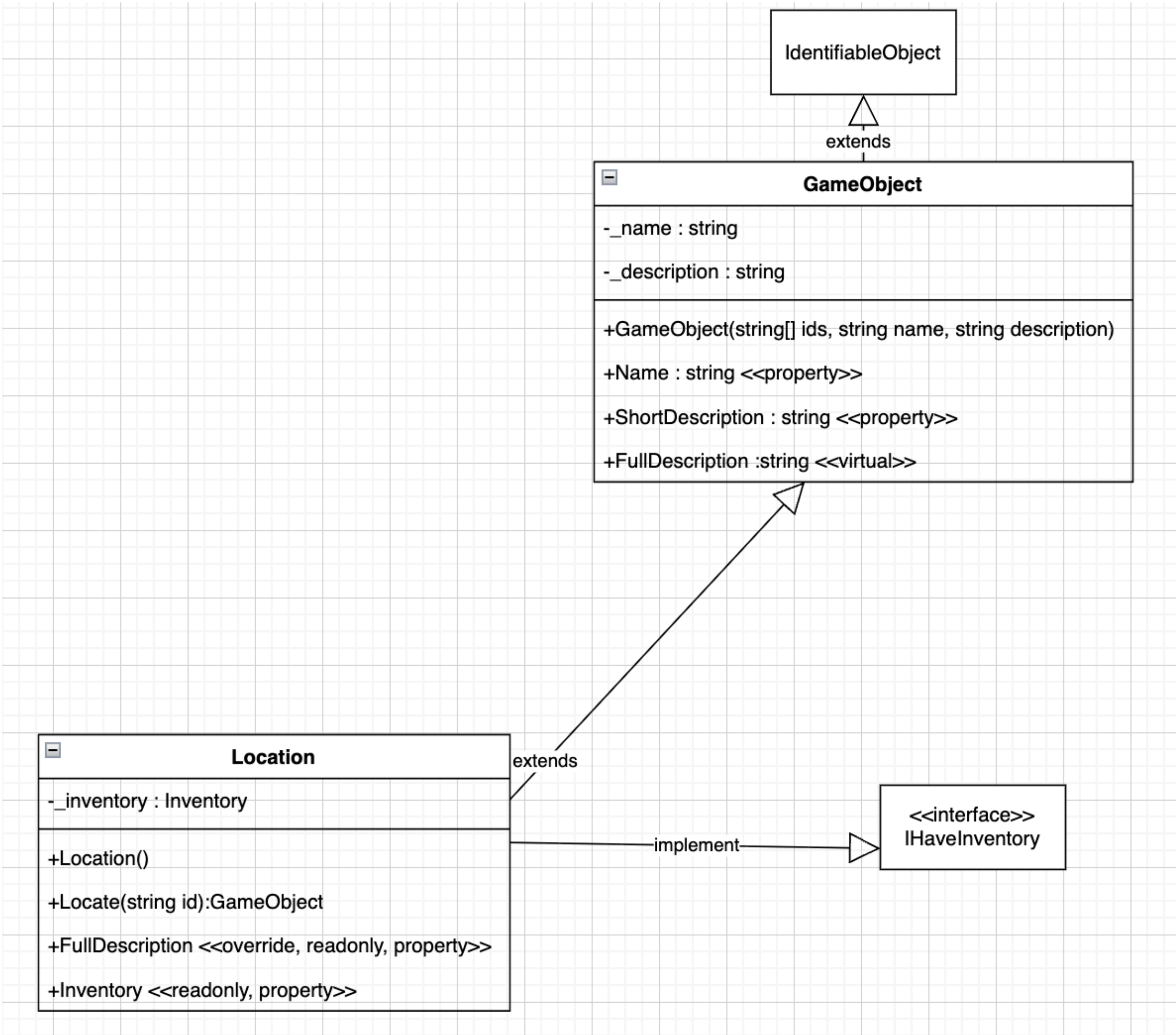
```
using SwinAdventure;
using NUnit.Framework.Legacy;
namespace UnitTest;

[TestFixture]
public class TestLocation
{
    private Location room;
    private Item gem;
    private Player player;

    [SetUp]
    public void SetUp()
    {
        room = new Location("hall", "a large empty hall");
        gem = new Item(new string[] { "Gem", "a gem" }, "A beautiful gem", "A beautiful gem that
you found in the chest");
        player = new("Fred", "a mighty adventurer");
        player.Location = room;
        room.Inventory.Put(gem);
```

```
    }
    [Test]
    public void Location_IdentifiesItself()
    {
        ClassicAssert.IsTrue(room.AreYou("hall"));
        ClassicAssert.AreEqual(room, room.Locate("hall"));
    }
    [Test]
    public void Location_CanLocateItemInInventory()
    {
        GameObject found = room.Locate("gem");
        ClassicAssert.IsNotNull(found);
        ClassicAssert.AreEqual(gem, found);
    }
    [Test]
    public void Player_CanLocateItemInTheirLocation()
    {
        GameObject found = player.Locate("gem");
        ClassicAssert.IsNotNull(found);
        ClassicAssert.AreEqual(gem, found);
    }

}
```

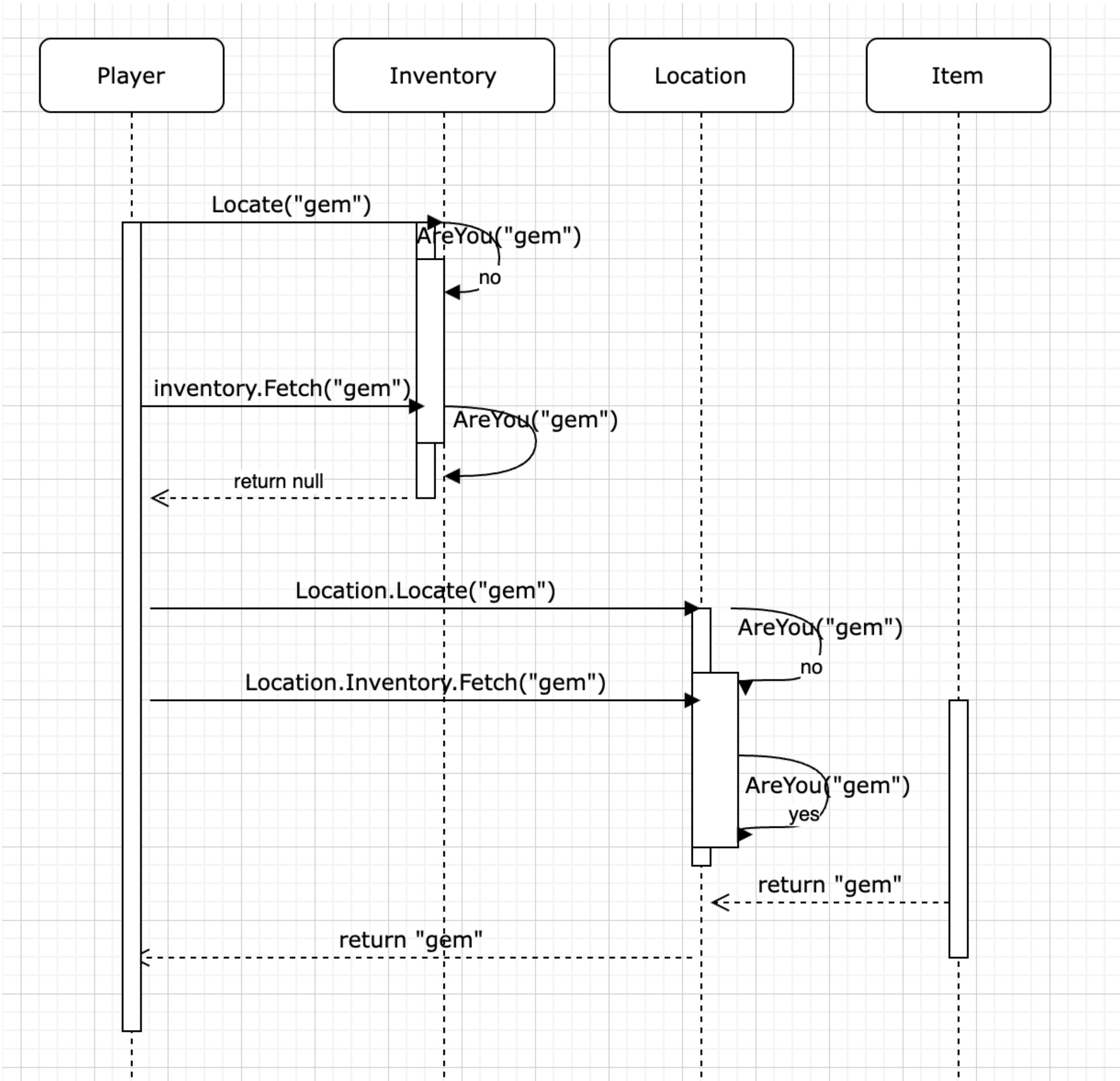
TestCases


Location_IdentifiesItself TestIdentifiableObject (net9.0) › UnitTest › ...
Location_CanLocateItemInInventory TestIdentifiableObject (net9.0...
Player_CanLocateItemInTheirLocation TestIdentifiableObject (net9....
TestAreYou TestIdentifiableObject (net9.0) › UnitTest › ...
TestAdd
TestIdentifiableObject (net9.0) › UnitTest › IdentifiableObjectTest (Passed), in 0.0ms
TestBagLocatesItems TestIdentifiableObject (net9.0) › UnitTest › Te...
TestBagLocatesItself TestIdentifiableObject (net9.0) › UnitTest › Tes...
TestBagLocatesNothing TestIdentifiableObject (net9.0) › UnitTest › ...
TestBagFullDescription TestIdentifiableObject (net9.0) › UnitTest › T...
TestBagInBag TestIdentifiableObject (net9.0) › UnitTest › TestBag: 0....
TestBagInBagWithPrivilegedItem TestIdentifiableObject (net9.0) › ...
TestCaseSEnsitive TestIdentifiableObject (net9.0) › UnitTest › Identif...
TestFindItem TestIdentifiableObject (net9.0) › UnitTest › InventoryTe...
TestFetchItem TestIdentifiableObject (net9.0) › UnitTest › InventoryT...
TestFirstID TestIdentifiableObject (net9.0) › UnitTest › IdentifiableObj...
TestFirstIDWithNoIDs TestIdentifiableObject (net9.0) › UnitTest › Ide...
TestItemIsIdentifiable TestIdentifiableObject (net9.0) › UnitTest › Ite...
TestFullDescription TestIdentifiableObject (net9.0) › UnitTest › ItemT...
TestItemList TestIdentifiableObject (net9.0) › UnitTest › InventoryTes...
TestLookAtMe TestIdentifiableObject (net9.0) › UnitTest › TestLookC...
TestLookAtGem TestIdentifiableObject (net9.0) › UnitTest › TestLook...
TestLookAtUnk TestIdentifiableObject (net9.0) › UnitTest › TestLook...
TestLookAtGemInMe TestIdentifiableObject (net9.0) › UnitTest › Tes...
TestLookAtGemInBag TestIdentifiableObject (net9.0) › UnitTest › Te...
TestLookAtGemInNoBag TestIdentifiableObject (net9.0) › UnitTest › ...
TestLookAtNoGemInBag TestIdentifiableObject (net9.0) › UnitTest › ...
TestInvalidLook TestIdentifiableObject (net9.0) › UnitTest › TestLook...
TestNoItemFind TestIdentifiableObject (net9.0) › UnitTest › Inventory...
TestNotAreYou TestIdentifiableObject (net9.0) › UnitTest › Identifiabl...
TestPlayerIsIdentifiable TestIdentifiableObject (net9.0) › UnitTest › ...
TestPlayerLocatesItems TestIdentifiableObject (net9.0) › UnitTest › ...
TestPlayerLocatesItself TestIdentifiableObject (net9.0) › UnitTest › ...
TestPlayerLocatesNothing TestIdentifiableObject (net9.0) › UnitTest...
TestPlayerFullDescription TestIdentifiableObject (net9.0) › UnitTest ...
TestShortDescription TestIdentifiableObject (net9.0) › UnitTest › Ite...
TestTakeItem TestIdentifiableObject (net9.0) › UnitTest › InventoryTe...
public Player(st
{
_inventory =
_location =
}
9 references
public GameObje
{
if (AreYou(
GameObject
if (item !=
if (Locatio
return null;
}
9 references
public override
{
get
{
return
}
}
11 references
public Inventory
{
get { return
}
4 references
PROBLEMS OUTPUT TERMINAL ...
Determining projects to restore...
All projects are up-to-date for re
IdentifiableObject -> /Users/khant
TestIdentifiableObject -> /Users/k
* Terminal will be reused by tasks
* Executing task: dotnet: build /U
dotnet build /Users/khantthuaung/Des
leloggerparameters:NoSummary /p:Conf
C# extension build result service is
Determining projects to restore...

UML Diagram



Sequence Diagram


Player
Inventory
Location
Item
Locate("gem")
AreYou("gem")
no
inventory.Fetch("gem")
AreYou("gem")
return null
Location.Locate("gem")
AreYou("gem")
no
Location.Inventory.Fetch("gem")
AreYou("gem")
yes
return "gem"
return "gem"